

Gaurav Girish Rathod

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [LeetCode](#) | [Codeforces](#)

PROFESSIONAL SUMMARY

- Full-stack and AI systems builder with strong CS fundamentals and a track record of 0-to-1 prototyping. Experienced in shipping responsive web applications (React, Next.js, TypeScript), backend services (Node.js, Express, REST APIs), and production GenAI pipelines utilizing LLMs, embeddings, and vector indexing. Proven cross-functional ownership leading 25+ member teams.

EDUCATION

- Indian Institute of Technology, Mandi** Himachal Pradesh, India
B.Tech in Civil Engineering; CGPA: 8.34 August 2024 – Present
– **Relevant Coursework:** Programming and Data Structures, Applied Electronics, Machine Learning, Deep Learning and Applications, Computing and Data Science, Design of Algorithms, Computer Networks, Database Management Systems, Operating Systems, Probability and Statistics, Linear Algebra
- Queen Mary** Mumbai, Maharashtra, India
Higher Secondary Certificate (HSC); Percentage: 85.5% 2022 – 2024

TECHNICAL SKILLS

- Full Stack & Web:** TypeScript, JavaScript (ES6+), React, Next.js, Node.js, Express.js, HTML5/CSS3, RESTful APIs
- GenAI & Machine Learning:** LLMs, Vector Databases, Embeddings, RAG, MCP, PyTorch, OpenCV, HuggingFace
- Core Languages:** C, C++17, Python, SQL, Linux/Bash
- Databases & Cloud:** PostgreSQL, SQL Schema Design, Neo4j, Azure CosmosDB, Docker, Git, CI/CD, AWS, Vercel

TECHNICAL PROJECTS

- Robust Deepfake Detection Under Domain Shift** [GitHub](#) | [Live](#)
PyTorch, DINOv2, Optimal Transport (FGW), SRM, Python 2026
– **Dual-Branch Architecture:** Coupled a frozen 307M-parameter DINOv2 ViT-L/14 with a 30-kernel SRM frequency stream, projecting 1024-d patch tokens and 128-d noise vectors into an 1153-d multimodal fusion head.
– **SSL & Graph Matching:** Built a 32-prototype real-face graph bank; deployed an FGW module with a 20-iteration Sinkhorn solver to match patch graphs ($K = 3$) and yield 224×224 anomaly heatmaps.
– **Benchmark Outperformance:** Surpassed published XceptionNet baselines on WildDeepfake across all metrics: lifted Accuracy from 72.3% $\rightarrow$ **76.5%**, Recall from 48.0% $\rightarrow$ **74.0%** (+26.0%), and F1-score from 0.59 $\rightarrow$ **0.76**.
– **Cross-Domain Generalization:** Attained **0.884 cross-domain AUC** (vs. 0.831 baseline, +5.3%) and **0.946 validation AUC** (88.7% accuracy), mitigating performance degradation under heavy compression across 140k+ test samples.
- Process Discovery & HITL Automation Engine** [GitHub](#)
Python, Playwright, Scikit-Learn, PaddleOCR, Process Mining 2026
– **Work-Unit Recovery:** Built a temporal boundary-detection pipeline on 183k+ operation logs using HistGradientBoosting, achieving **0.905 F1** and consolidating 540 unlabelled work units across 15 production sessions with 1.10s mean error.
– **Multimodal Process Mining:** Engineered an isolated PaddleOCR extraction pipeline (> 0.97 confidence) over 1.1k+ UI captures, categorizing 5 operational families and isolating 209 target workflow units consuming 35.8% of total back-office runtime.
– **HITL Automation Architecture:** Architected a parameterized Playwright browser automation suite featuring safe-stop invariants and strict human-in-the-loop gates; validated across 16/16 test scenarios with zero unauthorized transactions.
- Path-Finder: Algorithmic Graph Traversal Engine** [GitHub](#) | [Live](#)
React, JavaScript, CSS3, Graph Algorithms, Web Performance 2026
– **Traversal Engine:** Implemented interactive visualizations for A*, Dijkstra, BFS, and DFS on dynamically weighted grid topologies, rendering 2,500+ nodes under 16ms animation frames.
– **State Optimization:** Decoupled path calculation from UI re-renders using custom heuristic queues, mitigating redundant state transitions and maintaining seamless 60 FPS visual execution.
– **Modular Tooling:** Built recursive maze generation and state-export tools, cutting map setup time by 70%.

LEADERSHIP EXPERIENCE

- Sponsorship Head – Xpecto '26, IIT Mandi, Himachal Pradesh** September 2025 – March 2026
– **0-to-1 Ownership:** Led a 25-member cross-functional team, spearheading partner outreach and negotiating contracts to secure **INR 20+ Lakh** from 20+ corporate and public entities.
- Team Lead – Design Practicum, IIT Mandi, Himachal Pradesh** August 2025 – December 2025
– **Product Prototyping:** Spearheaded end-to-end development of an IoT neonatal warmer from concept to functional prototype, reducing bill of materials to **INR 8,000** vs. the INR 1 Lakh commercial baseline.

HONORS & ACHIEVEMENTS

- Competitive Programming:** Solved 500+ algorithmic problems across LeetCode and Codeforces; Codeforces rating: 1200+.
- Hackathon Finalist:** 1st Runner-Up in Deep Learning Hackathon HACK60 for developing an end-to-end computer vision solution.
- Inter-IIT Civil Conclave:** Won **3rd Position** pan-IIT in the Sustainable City Planning Case Study at IIT Roorkee.